

The Silent Tax: Typo-Robustness Analysis (ARC)

DeepSeek-R1-Distill-Qwen-7B

All tables are *answered-only*: traces that hit the token cap before writing a final answer are excluded from the metrics. Configs are named `typo{rate}_real{ratio}`: *rate* = % of words corrupted, *real* = % of typos that form real words. Baseline is the clean input.

Accuracy and Flips

Table 1: Accuracy per config. `n` = questions; `n_answered` = questions that produced a final answer (not truncated at the token cap). `acc_answered` = correct / `n_answered`; `acc_strict` = correct / `n` (truncated counted as wrong); `ci_lo/ci_hi` = 95% bootstrap CI on `acc_strict`.

config	n	n_answered	acc_answered	acc_strict	ci_lo	ci_hi
clean	500	485	0.8474	0.822	0.786	0.856
typo25_real10	500	487	0.8645	0.842	0.808	0.874
typo25_real40	500	486	0.856	0.832	0.798	0.862
typo25_real70	500	489	0.82	0.802	0.764	0.836
typo50_real10	500	487	0.8193	0.798	0.762	0.832
typo50_real40	500	485	0.8247	0.8	0.764	0.834
typo50_real70	500	484	0.8244	0.798	0.76	0.834
typo75_real10	500	488	0.8074	0.788	0.754	0.822
typo75_real40	500	485	0.7732	0.75	0.71	0.788
typo75_real70	500	486	0.7469	0.726	0.686	0.766

Table 2: Answered-only flips vs clean, paired by question over `n_both` (answered in both conditions). `r2w` / `w2r` = questions correct on clean but wrong on this config / wrong on clean but correct on this config; `r2w_ratio` = `r2w` / `n_both`, `w2r_ratio` = `w2r` / `n_both`; `net_flip_ratio` = `r2w_ratio` - `w2r_ratio` = (`r2w` - `w2r`) / `n_both`, the net rate at which this config worsens answers relative to clean.

config	n_both	r2w	w2r	r2w_ratio	w2r_ratio	net_flip_ratio	mcnemar_b	mcnemar_c	mcnemar_stat	mcnemar_p
typo25_real10	473	24	31	0.0507	0.0655	-0.0148	24	31	0.655	0.41849223
typo25_real40	472	28	33	0.0593	0.0699	-0.0106	28	33	0.262	0.60854777
typo25_real70	474	42	26	0.0886	0.0549	0.0338	42	26	3.309	0.06890881
typo50_real10	472	41	27	0.0869	0.0572	0.0297	41	27	2.485	0.11491486
typo50_real40	471	42	28	0.0892	0.0594	0.0297	42	28	2.414	0.1202328
typo50_real70	469	45	29	0.0959	0.0618	0.0341	45	29	3.041	0.08120887
typo75_real10	475	55	36	0.1158	0.0758	0.04	55	36	3.56	0.05917207
typo75_real40	471	63	29	0.1338	0.0616	0.0722	63	29	11.837	0.00058067
typo75_real70	472	75	26	0.1589	0.0551	0.1038	75	26	22.812	1.79e-06

McNemar test (used in the flips table above).

- **What it is:** a paired significance test for before/after binary outcomes. It looks only at the questions that *changed* (the discordant pairs: *b* = right→wrong, *c* = wrong→right) and ignores those that stayed the same.

- **Null hypothesis:** typos break correct answers as often as they fix wrong ones ($b = c$, i.e. no net effect).
- **Statistic:** $(|b - c| - 1)^2 / (b + c)$ (column `mcnemar_stat`); `mcnemar_p` is its chi-square p -value with 1 degree of freedom.
- **When we reject it:** when `mcnemar_p` < 0.05 — the right→wrong vs wrong→right imbalance is too large to be chance.

Reasoning Length

Table 3: Absolute generated tokens per config (answered-only).

config	n_answered	mean_tok	median_tok	p90_tok	max_tok
clean	485	1172.1	842	2238	8512
typo25_real10	487	1221.9	869	2400	8753
typo25_real40	486	1245.6	872.5	2588	8684
typo25_real70	489	1233.8	871	2680	7092
typo50_real10	487	1298.2	936	2615	9068
typo50_real40	485	1357.4	879	2935	9110
typo50_real70	484	1329.2	922.0	2752	8588
typo75_real10	488	1380.7	963.0	2752	7480
typo75_real40	485	1412.8	930	3084	7686
typo75_real70	486	1452.9	998.0	3154	11834

Table 4: Mean generated tokens for correct vs wrong answers. `wrong_over_correct` = `tok_wrong` / `tok_correct` (mean tokens of wrong answers divided by mean tokens of correct answers).

config	tok_correct	tok_wrong	wrong_over_correct
clean	1024.9	1989.3	1.9409
typo25_real10	1120.3	1870.4	1.6695
typo25_real40	1114.3	2025.7	1.8179
typo25_real70	1066.2	1997.5	1.8735
typo50_real10	1184.4	1814.5	1.532
typo50_real40	1206.3	2068.6	1.7148
typo50_real70	1179.6	2031.6	1.7222
typo75_real10	1231.6	2005.7	1.6286
typo75_real40	1262.6	1924.8	1.5244
typo75_real70	1262.6	2014.7	1.5957

Self-doubt

The self-doubt markers are curated words grouped into two *banks*, each matched as a case-insensitive whole-word pattern in the reasoning text. Density is matches per 1000 reasoning words.

- **second_guess** — the proposal’s canonical self-correction words plus direct reversals: *wait, hmm, actually, reconsider, but wait, hold on, on second thought, let me recheck, double-check.*
- **uncertainty** — hedging / “not certain” words: *maybe, perhaps, possibly, might be, could be, not sure, I guess, presumably, or maybe / alternatively, unclear / confused.*

Example reasoning: “*Wait, maybe the total is 12... hmm, actually let me recheck*” contributes *wait*, *hmm*, *actually* (second_guess) and *maybe* (uncertainty). A separate *typo-noticing* bank (*typo*, *misspelled*, *doesn't make sense*, ...) is treated as repair behaviour, not self-doubt.

Table 5: Doubt-marker density per config: second_guess and uncertainty, per 1k words and per trace.

config	n_answered	second_guess_per_1k	second_guess_per_trace	uncertainty_per_1k	uncertainty_per_trace	total_per_1k	total_per_trace
clean	485	11.332	9.136	13.995	11.282	25.327	20.419
typo25_real10	487	11.807	9.916	14.511	12.187	26.318	22.103
typo25_real40	486	11.012	9.377	13.399	11.409	24.411	20.786
typo25_real70	489	10.85	9.245	14.574	12.419	25.424	21.665
typo50_real10	487	10.975	9.735	14.69	13.031	25.665	22.766
typo50_real40	485	11.737	10.969	16.267	15.202	28.004	26.171
typo50_real70	484	11.101	10.225	16.141	14.868	27.242	25.093
typo75_real10	488	11.545	10.969	17.063	16.213	28.608	27.182
typo75_real40	485	11.337	11.027	18.354	17.852	29.692	28.878
typo75_real70	486	11.642	11.675	19.197	19.251	30.839	30.926

Table 6: Per-marker density (markers per 1000 reasoning words): clean vs pooled-typo. discrimination = typo minus clean (larger = more typo-responsive). All bank markers, ranked by discrimination.

marker	category	clean_per_1k	typo_per_1k	discrimination
or maybe/alt	uncertainty	1.941	2.624	0.683
maybe	uncertainty	6.98	7.487	0.507
perhaps	uncertainty	1.476	1.9	0.425
unclear/confused	uncertainty	0.604	0.806	0.202
wait	second_guess	6.092	6.264	0.172
could be	uncertainty	0.468	0.548	0.08
might be	uncertainty	0.844	0.923	0.079
not sure	uncertainty	1.325	1.383	0.058
not entirely	uncertainty	0.266	0.319	0.053
hold on	second_guess	0.217	0.258	0.041
I guess	uncertainty	0.043	0.062	0.018
I assume	uncertainty	0.005	0.023	0.017
let me recheck	second_guess	0.164	0.174	0.011
possibly	uncertainty	0.036	0.046	0.01
on second thought	second_guess	0.0	0.0	0.0
presumably	uncertainty	0.008	0.006	-0.002
reconsider	second_guess	0.015	0.01	-0.005
double-check	second_guess	0.281	0.267	-0.015
wait/actually no	second_guess	0.491	0.476	-0.015
but wait	second_guess	0.969	0.931	-0.039
actually	second_guess	0.772	0.703	-0.07
hmm	second_guess	2.33	2.259	-0.071

Table 7: Doubt density (markers per 1000 reasoning words) split by final outcome, per category (2g = second.guess, un = uncertainty, tot = both). For a group, density = total markers in that group / total reasoning words in that group x 1000. *_correct = over correct-answer traces; *_wrong = over wrong-answer traces; *_wrong_over_correct = _wrong / _correct.

config	2g_correct	2g_wrong	2g_wrong_over_correct	un_correct	un_wrong	un_wrong_over_correct	tot_correct	tot_wrong	tot_wrong_over_correct
clean	10.713	12.97	1.211	11.844	19.679	1.661	22.557	32.649	1.447
typo25_real10	11.063	14.54	1.314	12.939	20.29	1.568	24.002	34.83	1.451
typo25_real40	10.774	11.76	1.092	11.436	19.581	1.712	22.21	31.341	1.411
typo25_real70	10.276	12.145	1.182	11.322	21.921	1.936	21.598	34.066	1.577
typo50_real10	10.716	11.753	1.097	13.051	19.626	1.504	23.767	31.379	1.32
typo50_real40	11.612	12.065	1.039	12.968	24.922	1.922	24.581	36.987	1.505
typo50_real70	10.664	12.242	1.148	13.617	22.733	1.669	24.281	34.975	1.44
typo75_real10	11.54	11.557	1.002	14.991	22.133	1.476	26.531	33.69	1.27
typo75_real40	11.352	11.307	0.996	16.328	22.601	1.384	27.679	33.908	1.225
typo75_real70	11.637	11.651	1.001	16.419	24.121	1.469	28.056	35.772	1.275

LLM judge. The proposal specifies a second, non-lexical measure for both dimensions: a separate model with a fixed, strict prompt. The judge is a different model from the one under test, called at temperature 0 on a seeded random subsample of answered traces per config. It is shown the corrupted-word list (*original* $\rightarrow$ *as shown*), the question, and the reasoning trace (head+tail window). It returns two scalar ratings and evidence: *self_doubt_score* from 0–10, where 0 means no second-guessing and 10 means pervasive uncertainty or looping; and *repair_understanding_score* from 0–5, where 0 means the question’s meaning is fully understood and 5 means the model is fundamentally lost because of the corruption. The judge also returns a one-line justification and verbatim evidence quotes; quotes are checked against the reasoning trace. The judge is a subsample measure, while the lexical tables remain the full-data result.

Table 8: New scalar LLM-judge analysis per config. mean/median *self_doubt_score* use the 0-10 scale; mean/median *repair_understanding_score* use the 0-5 scale; threshold columns show the share of traces with substantial doubt ($i=5$) or substantial loss of meaning ($i=3$).

config	n	mean_self_doubt	median_self_doubt	frac_self_doubt_ge5	mean_repair_understanding	median_repair_understanding	frac_repair_understanding_ge3	evidence_self_doubt_frac	evidence_repair_frac
clean	443	4.693	5	0.5892	0.929	0	0.0045	0.9503	0.9955
typo25_real10	481	5.04	5	0.657	0.8	1	0.0395	0.9397	0.842
typo25_real40	481	5.077	5	0.6528	0.952	1	0.0665	0.9252	0.8586
typo25_real70	486	5.239	5	0.6831	1.152	1	0.0967	0.9444	0.8292
typo50_real10	480	5.371	6	0.7	0.89	1	0.0792	0.9354	0.8812
typo50_real40	454	5.185	5	0.674	0.987	1	0.0947	0.9339	0.848
typo50_real70	481	5.366	6	0.684	1.204	1	0.1289	0.9231	0.8337
typo75_real10	481	5.572	6	0.7277	1.123	1	0.1143	0.9023	0.8233
typo75_real40	477	5.556	6	0.7191	1.354	1	0.1824	0.9182	0.7925
typo75_real70	480	5.754	6	0.7354	1.587	1	0.2292	0.9042	0.75

Table 9: New scalar LLM-judge scores split by final answer outcome. The correct/wrong columns are means over traces whose final answer was correct or wrong.

config	n	n_correct	n_wrong	mean_doubt_correct	mean_doubt_wrong	mean_repair_correct	mean_repair_wrong
clean	443	375	68	4.333	6.676	0.021	0.074
typo25_real10	481	417	64	4.772	6.781	0.707	1.406
typo25_real40	481	411	70	4.793	6.743	0.839	1.614
typo25_real70	486	398	88	4.842	7.034	1.008	1.807
typo50_real10	480	394	86	5.081	6.698	0.759	1.488
typo50_real40	454	379	75	4.844	6.907	0.892	1.467
typo50_real70	481	397	84	4.99	7.143	1.065	1.857
typo75_real10	481	390	91	5.259	6.912	0.985	1.714
typo75_real40	477	368	109	5.217	6.697	1.212	1.835
typo75_real70	480	359	121	5.312	7.066	1.379	2.207
__all__	4744	3888	856	4.941	6.877	0.883	1.617

Table 10: New scalar LLM-judge scores pooled by real-word typo ratio, showing whether higher real-word corruption is associated with more doubt or loss of understanding.

real_ratio	n	mean_self_doubt	mean_repair_understanding	frac_self_doubt_ge5	frac_repair_understanding_ge3
10	1442	5.327	0.938	0.6949	0.0777
40	1412	5.273	1.099	0.682	0.1147
70	1447	5.452	1.314	0.7008	0.1513

Repair Behaviour

Table 11: How each corrupted word was handled, per config (categories defined below the table).

config	n_corrupt_words	silent_fix_n	flagged_n	misread_n	not_used_n	silent_fix_frac	flagged_frac	misread_frac	not_used_frac
typo25_real10	1964	1572	114	41	237	0.8004	0.058	0.0209	0.1207
typo25_real40	1981	1481	180	61	259	0.7476	0.0909	0.0308	0.1307
typo25_real70	1988	1384	243	105	256	0.6962	0.1222	0.0528	0.1288
typo50_real10	3932	3023	317	100	492	0.7688	0.0806	0.0254	0.1251
typo50_real40	3927	2821	459	147	500	0.7184	0.1169	0.0374	0.1273
typo50_real70	3887	2594	551	208	534	0.6674	0.1418	0.0535	0.1374
typo75_real10	5847	4154	784	196	713	0.7104	0.1341	0.0335	0.1219
typo75_real40	5777	3802	944	291	740	0.6581	0.1634	0.0504	0.1281
typo75_real70	5769	3429	1174	398	768	0.5944	0.2035	0.069	0.1331

A *corrupted word* is a word that received a typo — it differs between the clean and typo question (e.g. *sum* → *sun*). We recover these by diffing the two questions, then place each into one bucket by which form appears in the reasoning:

- **silent_fix** — only the original word (*sum*) appears: a quiet recovery.
- **flagged** — both forms appear (*sum* and *sun*): noticed and fixed.
- **misread** — only the corrupted form (*sun*) appears: read as the wrong word.
- **not_used** — neither form appears in the reasoning.

Columns ***_n** are counts; ***_frac** are the share of **n_corrupt_words**.

Table 12: Word handling pooled by real-word ratio: the non-word vs real-word contrast.

real_ratio	n_corrupt_words	silent_fix_frac	flagged_frac	misread_frac	not_used_frac
10	11743	0.745	0.1035	0.0287	0.1228
40	11685	0.6935	0.1355	0.0427	0.1283
70	11644	0.6361	0.169	0.0611	0.1338

Table 13: Misread rate split by final correctness. A typo’d word is ‘misread’ when the reasoning uses only its corrupted form (e.g. sun instead of sum). Within each outcome group, misread rate = (typo’d words that were misread) / (all typo’d words examined), pooled over the group. misread_correct = over problems answered correctly; misread_wrong = over problems answered wrong; wrong_over_correct = misread_wrong / misread_correct.

config	misread_correct	misread_wrong	wrong_over_correct
typo25_real10	0.0178	0.0391	2.196
typo25_real40	0.0269	0.0521	1.939
typo25_real70	0.0501	0.0641	1.28
typo50_real10	0.0201	0.0485	2.418
typo50_real40	0.0342	0.0514	1.504
typo50_real70	0.0515	0.0615	1.193
typo75_real10	0.0293	0.0516	1.76
typo75_real40	0.0466	0.062	1.328
typo75_real70	0.0606	0.0931	1.535

Real-word Effect

This section isolates the effect of typo *kind* (real-word vs non-word), holding the question, corrupted positions, and rate fixed — something the vs-clean accuracy section cannot do. Three views:

- a **controlled paired** comparison of low-real vs high-real variants on the *same* questions, with a McNemar test (paired table);
- a **per-typo logistic regression** comparing the marginal harm of one real-word vs one non-word typo (logit table);
- whether wrong answers **flag the corruption** more or less as the real-word share rises (silent-failure table).

Read the signs, deltas and p -values in the tables below for this dataset’s result.

Table 14: Controlled paired comparison of two TYPO variants against EACH OTHER (baseline is the low-real variant, NOT clean – so this is not the vs-clean accuracy table). Within a fixed rate the same questions are corrupted in the same positions, only the typo KIND differs. Restricted to questions answered in both variants. acc_low_real / acc_high_real = accuracy on the lower / higher real-word variant; delta_acc = acc_high_real - acc_low_real; r2w = questions correct in the low-real variant but wrong in the high-real one; w2r = the reverse; mcnemar_p = McNemar test on that paired change. Negative delta with r2w $\hat{>}$ w2r means real-word typos are more harmful than the non-word typos they replaced.

rate	compare	n_both	acc_low_real	acc_high_real	delta_acc	r2w	w2r	mcnemar_p
25	real10_to_real70	476	0.8655	0.8277	-0.0378	46	28	0.04813101
25	real10_to_real40	474	0.865	0.8608	-0.0042	32	30	0.8989403
25	real40_to_real70	477	0.8616	0.8239	-0.0377	40	22	0.03085
50	real10_to_real70	472	0.8263	0.8326	0.0064	38	41	0.8219656
50	real10_to_real40	473	0.8288	0.8288	0.0	35	35	0.90486113
50	real40_to_real70	472	0.8369	0.8284	-0.0085	39	35	0.72728232
75	real10_to_real70	475	0.8126	0.7453	-0.0674	60	28	0.00095108
75	real10_to_real40	474	0.8143	0.7743	-0.0401	54	35	0.05639171
75	real40_to_real70	471	0.7834	0.7495	-0.034	60	44	0.141326

Table 15: Logistic regression $\text{correct} \sim \text{num_real} + \text{num_nonword}$ over all answered typo traces ($\text{num_real} / \text{num_nonword}$ = count of real-word / non-word typos in the problem). coef_* = change in log-odds of a correct answer per one added typo of that kind (more negative = more harmful); $\text{odds_*} = \exp(\text{coef})$; intercept is the model constant. n = number of traces fit.

n	coef_num_real	coef_num_nonword	odds_real	odds_nonword	intercept
4377	-0.04182	-0.00784	0.959	0.9922	1.67003

Table 16: Among WRONG answers only, does the model flag the corruption, by real ratio? $\text{explicit_notice_frac}$ = fraction of wrong-answer traces whose reasoning uses strong typo-flagging language (typo / misspelled / doesn't make sense / ...); $\text{silent_frac} = 1 - \text{explicit_notice_frac}$. The proposal predicts real-word failures are quieter ($\text{explicit_notice_frac}$ should fall as real rises).

real_ratio	n_wrong	explicit_notice_frac	silent_frac
10	248	0.3548	0.6452
40	265	0.3811	0.6189
70	296	0.3986	0.6014

Lexical Grid (overview)

Table 17: All four marker families (second_guess , uncertainty , typo_noticing , repair_words) across configs.

config	second_guess	uncertainty	typo_noticing	repair_words
clean	11.332	13.995	1.084	0.801
typo25_real10	11.807	14.511	1.416	0.922
typo25_real40	11.012	13.399	1.51	1.037
typo25_real70	10.85	14.574	1.555	1.056
typo50_real10	10.975	14.69	1.588	1.012
typo50_real40	11.737	16.267	1.78	1.143
typo50_real70	11.101	16.141	1.736	1.099
typo75_real10	11.545	17.063	1.926	1.199
typo75_real40	11.337	18.354	2.181	1.376
typo75_real70	11.642	19.197	2.407	1.524

Spellcheck Recovery